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Comparison of Post-Hemorrhoid Surgery Pain Profiles in Older and Younger Adults



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ABSTRACT

Introduction: Older and younger adults are offered similar analgesic options after hemorrhoid surgery (HS), but the differences in pain between the two populations are unknown. This study aims to compare postoperative pain outcomes after HS in older and younger individuals.

Methods: This is a retrospective analysis of electronic medical records of patients who underwent HS between 2018 and 2023. Patients were excluded if additional anorectal procedures were performed at the time of HS. Data related to pain-related outcomes were compiled: (1) need for narcotic prescription refills; (2) documentation of a pain-related phone call within 30 d; (3) urgent postoperative office visit before regular scheduled follow-up; and (4) pain-related postoperative emergency department visits. Associations between age and pain-related outcomes were tested using Fisher's exact test, chi-square test, and covariate adjusted logistic regression modeling.

Results: There were a total of 249 patients, 60 older adults, and 189 younger adults. Compared to younger patients, older adults demonstrated a reduced frequency of pain-related phone calls (10.3 versus 32.1%, $P < 0.01$) and opioid refills (0 versus 14.4%, $P < 0.01$). After adjusting for confounders, older age remained inversely associated with pain-related postoperative phone calls (odds ratio = 0.25, 95% confidence interval = [0.1–0.6], $P = 0.003$).

Conclusions: Older adults had better pain outcomes after HS in comparison to younger patients. These findings suggest that the postoperative analgesic needs of older patients after HS are lower than those of younger patients. Decisions regarding opioid prescription in older adults recovering from HS should be tailored to avoid narcotic-related complications.

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Introduction

Pain management in older adults after hemorrhoid surgery (HS) has unique challenges due to the potential for adverse effects with opioids in this population. The American Geriatric Society recommends cautionary use of opioids in geriatric patients due to the increased risk of falls, dangerous pharmacological interactions, and delirium risk.¹ Despite these risks, analgesic guidelines specific to HS do not contain special provisions for older patients, and opioid prescription after HS is common among colorectal surgeons.^{2,3}

Older age is not only associated with the potential for opioid-related complications but has also been associated with decreased postsurgical pain.⁴ Though the underlying mechanism for this observation is not completely understood, aging is characterized by a decline in function for the nervous, musculoskeletal, and endocrine systems, each of which may impact nociception.⁵ Psychological and cultural differences across generations may also impact pain perception.⁶

Despite these previously reported observations, the potential difference in postoperative pain for older and younger adults following HS is not known. Defining the difference in postoperative pain between these groups in this particular context would be clinically important because of the possible impact on opioid prescription after HS. In this study, we aimed to compare postoperative pain following HS between older and younger adults and hypothesized that older patients would experience less postoperative pain when compared with younger patients.

Methods

Study design

A retrospective analysis was conducted, comparing two historical patient cohorts. The data used for this comparison were derived from the electronic medical record at two hospitals within a single hospital system. The study design was reviewed by the hospital institutional review board and granted exemption due to its retrospective nature.

Patient population and cohorts

Patients were included in the study if they underwent HS during a 5 y interval (2018 and 2023). Cases were initially selected based on Current Procedural Terminology codes, and codes for both excisional hemorrhoidectomy (EH) and transanal hemorrhoidal dearterialization (THD) were included. All surgeries were performed by a single surgeon. Patients were excluded if additional anorectal procedures were performed simultaneously at the time of HS, if they were inpatient status at the time of the operation, or if they had a follow-up interval of <3 mo duration. The interval of 3 mo was selected to allow enough passage of time to capture all primary and secondary outcomes. Patients with inflammatory bowel disease were not included in this study. Patients who met the inclusion and exclusion criteria were divided into two

groups based on age at the time of operation. Older age was defined as ≥ 65 y of age, and younger age was <65 y of age, in accordance with prior methods.⁷⁻¹²

Surgical procedures and perioperative analgesia

All patients undergoing HS, either EH or THD, received preoperative education about postoperative pain control. This included instructions for liberal Sitz bath usage, encouragement of round-the-clock over the counter medications Tylenol and Motrin, and a discussion about bowel regimen. Immediately before the onset of the surgical procedure, patients received a bilateral pudendal nerve block and a circumferential perianal injection with both 1% lidocaine and liposomal bupivacaine. Unless the patient had underlying renal failure, 15 mg of intravenous ketorolac was given during the procedure as well. The surgical technique for EH was performed using a Ligasure according to previously published descriptions, with some minor modifications in terms of patient positioning (generally in left lateral decubitus) and surgical instrumentation (lighted Hill retractor).¹³ Number of excisional sites varied for patients and was determined intraoperatively. Technique for THD followed previously published methods, with again some minor modifications.¹⁴ Patients undergoing THD had six arterial ligations in the same anatomic locations described by Gachabayov *et al.*, but the same ligation suture was not used for mucopexy. Mucopexy was performed more selectively with separate 2-0 Vicryl sutures, and the number of pexies was at the surgeon's discretion based on the appearance of each hemorrhoidal column. Postoperative prescriptions included lidocaine 5% topical ointment three times daily as needed and 10 pills of oxycodone 5 mg q 4-6 h as needed.

Outcomes of interest

The primary outcome of interest in this study was postoperative pain. This was characterized using four different pain-related outcomes: (1) recording in the electronic medical record of a narcotic prescription refill; (2) documentation of a pain-related phone call to the surgery office within 30 d; (3) urgent postoperative office visit before regular scheduled follow-up; and (4) pain-related postoperative emergency room or hospital admission. Secondary outcomes included postoperative complications that were unrelated to pain (bleeding, infection, urinary retention, incontinence, and stenosis) and also included recurrence of hemorrhoidal disease.

Statistical analyses

The two study cohorts were compared to each other with respect to the study variables. In addition to the outcomes defined above, these included basic demographics (age, sex, and race), medical comorbidities (hypertension, diabetes, smoking history, and chronic opioid dependence), surgical/procedural history (any previous HS, prior rubber band ligation, and prior anorectal surgery), presenting symptoms (bleeding, prolapse, pain, and Goligher classification), and baseline anorectal function (history of constipation, incontinence, and

diarrhea). Categorical variables were compared using chi-square, or in the case of low-frequency events (<10), the Fisher exact test. Continuous variables were evaluated using Student's t-test. Statistical testing was performed in SAS Version 9.4.1 (Cary, NC) and in R Studio version February 1, 2024 (Integrated Development Environment, Boston, MA).

Results

A total of 249 patients were identified in the study population, including 60 (24.1%) older adults and 189 (75.9%) younger adults. The average time to follow-up was 73.7 d, and the time interval for most of the pain-related outcomes was within the first 3 wk. Descriptive analysis (Table 1) revealed that, when compared with the younger adults, older adults had a higher frequency of female patients (58.3% versus 40%, $P = 0.02$) and a higher frequency of hypertension (61.7% versus 34.4%, $P < 0.01$). Preoperatively, younger adults presented more often with anal pain (70.9% versus 46.7%, $P = 0.001$) and less often with incontinence (2.6% versus 13.3%, $P = 0.003$). Younger adults underwent less THD procedures in comparison to older patients (22.2% versus 40%, $P = 0.01$). There was no statistically significant difference between older adult and younger adult groups in terms of chronic opioid use (1.7% versus 4.2%, $P = 0.55$) or the distribution of preoperative Goligher classification ($P = 0.55$). Older adults had similar rates of prior hemorrhoidectomy (10% versus 13.3%, $P = 0.17$), prior rubber band ligation (16.7% versus 23.3%, $P = 0.06$), and other anorectal procedures (11.7% versus 25%, $P = 0.53$) when compared with younger adults. There were no cases of prior stroke, neuropathy, dementia, or cognitive disorder in either group.

Bivariate analyses of postoperative outcomes are presented in Table 2. When compared with younger adults, older adults had a significantly lower frequency of pain-related phone calls (10 versus 31.7%, $P = 0.003$) and opioid prescription refills (0% versus 14.3%, $P = 0.004$). Older and younger adult groups displayed no statistically significant differences between urgent clinic visits (15% versus 15.3%, $P = 0.69$) and emergency department admissions (8.3% versus 11.6%, $P = 0.63$) related to pain after surgery. Scatter plots were also constructed to visualize the distribution of age in each of the pain-related outcomes (Supplementary Fig. 1A-D). These graphs showed that the majority of pain-related phone calls and all of the opioid refills were distributed to ages younger than 65. Emergency department admissions and early clinic visits did not appear to have a clear age-related cutoff. Older adults had a higher rate of postoperative incontinence (8.3% versus 2.1%, $P = 0.02$). Older adults had a higher percentage of postoperative constipation in comparison to younger adults; however, this was not statistically significant (23.3% versus 13.2%, $P = 0.07$). There were no statistically significant differences between older and younger groups when comparing postoperative bleeding (35% versus 31.7%, $P = 0.43$), diarrhea (3.3% versus 3.2%, $P = 0.31$), recurrence (5% versus 9.5%, $P = 0.07$), and reoperation (1.7% versus 3.2%, $P = 0.87$).

The difference in pain-related phone calls between age groups remained robust when tested using a logistic regression model and adjusting for potential confounders (Table 3). Older age remained inversely associated with pain-related postoperative

Table 1 – Baseline characteristics for study cohorts.

Characteristic	Older Age ≥65	Younger Age <65	P
N (%)	60 (24.1)	189 (75.9)	
Age, y; median (range)	72 (65-91)	49 (20-64)	<0.001
Gender			
Male	25 (41.7)	112 (59.3)	0.02
Female	35 (58.3)	77 (40.7)	
Race			
White	32 (53.3)	97 (51.3)	0.68
Black	25 (41.7)	69 (36.5)	
Other	3 (5)	23 (12.2)	
Medical history			
Diabetes	6 (10)	17 (9)	1
Hypertension	37 (61.7)	65 (34.4)	<0.01
Smoking	5 (8.3)	30 (15.9)	0.07
Chronic opioid use	1 (1.7)	8 (4.2)	0.55
Surgical history			
Prior hemorrhoidectomy	6 (10)	8 (13.3)	0.17
Prior rubber band ligation	10 (16.7)	14 (23.3)	0.06
Other anorectal	7 (11.7)	15 (25)	0.53
Goligher classification			
1	0 (0)	1 (0.5)	0.52
2	3 (5)	7 (3.7)	
3	49 (81.7)	141 (74.6)	
4	8 (13.3)	40 (21.2)	
Presenting symptoms*			
Rectal bleeding	48 (80)	151 (79.9)	1
Anal pain	28 (46.7)	134 (70.9)	0.001
Prolapse	33 (55)	119 (63)	0.34
Incontinence	8 (13.3)	5 (2.6)	0.003
Constipation	19 (31.7)	65 (34.4)	0.81
Diarrhea	2 (3.3)	20 (10.6)	0.14
Surgical method			
Excisional	36 (60)	147 (77.8)	0.01
THD	24 (40)	42 (22.2)	

SD = standard deviation.

*Individual patients may have had multiple symptoms.

phone calls (odds ratio = 0.25, 95% confidence interval = [0.1-0.6], $P = 0.003$). Other independent variables (sex, race, procedure type, Goligher classification, number of excised/pexied hemorrhoids, preoperative pain, and chronic opioid use) were not associated with pain-related phone calls. Age was also tested as a continuous variable in a separate model and remained significant (odds ratio = 0.978, confidence interval [0.959-0.998], $P = 0.03$). Opioid refill could not be modeled in regression analysis since there were no events in the older group.

Discussion

The physiology of human aging includes decreased ability to maintain homeostasis and decreased reserve in the recovery

Table 2 – Comparison of postoperative outcomes for younger and older patients following hemorrhoid surgery.

Characteristic	Older Age ≥65	Younger Age <65	P
N (%)	60 (24.1)	189 (75.9)	
Pain-related outcomes			
Pain-related phone call	6 (10)	60 (31.7)	0.003
Opioid prescription refill	0 (0)	27 (14.3)	0.004
Urgent clinic visit	9 (15)	29 (15.3)	0.69
Emergency department admission	5 (8.3)	22 (11.6)	0.63
Incontinence	5 (8.3)	4 (2.1)	0.02
Constipation	14 (23.3)	25 (13.2)	0.07
Bleeding	21 (35)	60 (31.7)	0.43
Diarrhea	2 (3.3)	6 (3.2)	0.31
Recurrence	3 (5)	18 (9.5)	0.07
Reoperation	1 (1.7)	6 (3.2)	0.87

Bold indicates statistical significance $P < 0.05$.

from physiologic stress, including that of surgical procedures.^{15,16} These changes have been linked to alterations in pain perception by decreasing sensitivity of afferent somatosensory nerves and declination of receptor density.¹⁷ In this study, we compared postoperative pain outcomes after HS in older and younger adults and reported several findings of interest. First, it was observed that older adults were less likely to include pain or discomfort as a presenting symptom for their hemorrhoidal disease. This suggests a potential

Table 3 – Association between age category (independent variable) and pain-related phone call (dependent variable) using logistic regression modeling adjusted for confounders.

Variable	OR	95% CI	P
Age (ref younger)			
Older	0.25	0.10-0.63	0.003
Sex (ref female)			
Male	1.4	0.75-2.5	0.31
Race (ref Black)			
White	1.0	0.54-2.0	0.94
Other	1.1	0.43-3.1	0.79
Procedure type (EH)			
THD	0.66	0.04-9.8	0.80
Goligher classification (ref 3)			
Class 4	1.5	0.70-3.0	0.31
Number of hemorrhoids excised or pexied	1.0	0.41-2.6	0.94
History of opioid use	1.0	0.23-4.5	0.95
History of anal pain	1.2	0.61-2.3	0.61

Bold indicates statistical significance $P < 0.05$.

OR = odds ratio; CI = confidence interval; EH = excisional hemorrhoidectomy.

difference in hemorrhoidal pathophysiology and/or pain perception for the anorectum in different age groups, even before surgery. Second, we found significant differences in postoperative pain metrics between the age groupings. Older patients did not refill opioid prescriptions at all, whereas younger patients did so at a rate of 14.3%. Older patients had 75% lower odds of placing a postoperative pain-related phone call to the surgeon in comparison to younger patients, after adjusting for confounders. Looking at age as a continuous variable, for each 1-y increase in age, the odds of making a pain phone call decreased by approximately 2.2%. Taken together, these observations suggest that older and younger patients experience anorectal pain differently both before and after HS. These findings support the use of age-specific analgesic recommendations for patients recovering from HS, with an emphasis on opioid avoidance in the older population.

Age-related differences in pain after HS have been reported previously in several other studies. One example is in follow-up data from the NoTamp trial, a multicenter randomized controlled trial from Germany, which randomized 725 patients to receive a tamponade dressing or not and found that the use of tamponade dressings led to higher pain scores.¹⁸ Mallmann et al. found in their follow up analysis, that younger age (<40 y) was a strong risk factor for postoperative pain after EH (all either Milligan-Morgan or Parks technique), along with the use of tamponade dressing.¹⁹ In another study by Yamamoto et al., authors reviewed retrospective outcomes from Milligan-Morgan hemorrhoidectomy in a cohort of 100 patients, and found that older patients (defined in this study as ≥75 y) reported lower pain scores in the first 48 h after surgery when compared with younger patients.²⁰ Our findings in the present study are in agreement with, and expand upon, these other studies by demonstrating a similar age-related difference in pain over a longer postoperative interval.

Reports from other types of surgery reveal similar age-related differences in postoperative pain (general, abdominal, orthopedic, and gynecological surgery).²¹⁻²³ A qualitative systematic review assessing predictors of postoperative pain found younger age to be among the highest ranking predictors of postoperative pain.²¹ Age-related differences have not only been observed acutely after surgery but also more chronically at longer intervals. A prospective study assessing chronic postsurgical pain after colorectal surgery (abdominal, not anorectal) found that younger age was associated with an increased frequency of chronic postsurgical pain at 3 and 6 mo.²² A prospective study designed to create a reliable prediction model for acute postoperative pain and pain at 3 mo after surgery found that female sex and younger age were positively correlated with increased acute postoperative pain.²³ The contributors to postoperative pain are notably complex; however, our findings are in alignment with the broader consensus across various surgical specialties that an inverse relationship between age and pain exists.

In our study, sex and procedure type were not found to be associated with pain on the final regression model. In general, the association between sex and postoperative pain remains an area of debate; however, there are data that support the hypothesis that female sex is a predictor of higher postoperative pain scores or opioid consumption.²⁴⁻²⁶ Regarding procedure type, it has been observed by others in published

reports that THD is a less painful alternative to the traditional EH for hemorrhoids.^{27–29} The older subgroup of our cohort had a higher frequency of both female sex and THD procedures in comparison to the younger subgroup. In light of this and due to the aforementioned published data, these two variables were viewed as potential confounders, and adjustment for both lends strength to the final regression result.

Our findings support the idea that adequate pain control in older patients after HS may be feasible with more limited narcotic dosing compared to younger patients. A standardized approach for this has not yet been adopted in the United States and prescribing patterns after anorectal surgery vary greatly.³ Reducing opioid prescription and consumption in older patients is an important objective for colorectal surgeons because of the potential for adverse events in this age group and because unused narcotic pills that result from overprescription can end up in the hands of children, relatives, or friends of these patients and subjected to unauthorized use. One promising opportunity for opioid reduction for all age groups is the use of enhanced recovery after surgery (ERAS) pathways.³⁰ A recent randomized control trial found that ERAS protocol utilization after anorectal surgery led to significant reductions in postoperative opioid use.³⁰ As anorectal ERAS pathways become more standardized, we recommend age-specific recommendations for opioid use postoperatively, with an emphasis on dose reduction.

While our study provides valuable insights into the postoperative pain experience of older and younger patients after HS, it is not without limitations. The nature of retrospective cohort studies restricts our study from determining causality and is subject to information, selection, and confounding biases. To mitigate the effect of these biases on our study, only patients with sufficient follow-up and recorded data were included in the initial sample. Patients with blatant confounders (concomitant anorectal surgery, inpatient status) were excluded from further stratification. Another limitation of the study was the lack of a more granular or numeric pain-related outcome, such as the visual analog scale. Although numerical pain scores were collected after surgery and during follow-up visits, the length of time between surgery and the first follow-up was inconsistent and unsuitable for analysis. Another limitation is that we also do not have information regarding the number of opioid pills consumed or left over. Differences between older and younger groups in our chosen pain-related study outcomes may be due to factors unrelated to physical pain. For example, psychological, cultural, and socioeconomic differences between older and younger patients may not only impact their pain experience but also their likelihood of contacting their physicians when experiencing pain.^{31–33} Older patients are also less likely to be familiar with modern-day communication technology and own cell phones, which may impede their ability to contact providers to report pain or request refills.³⁴ Lower health literacy has also been associated with lower utilization of health resources.³⁵

Conclusions

Older adults had better pain outcomes after HS in comparison to younger patients, specifically fewer pain-related phone calls and zero opioid refills. These findings suggest that the

postoperative analgesic needs of older patients after HS are lower than those of younger patients. With the increased risk of adverse effects of using opioids in elderly patients and the ongoing opioid crisis, there is a motive to decrease opioid prescriptions after surgery when possible. Decisions regarding opioid prescription in older adults recovering from HS should be made in a tailored manner to avoid narcotic-related complications and reduce the supply of narcotics for nonmedical use.

Supplementary Materials

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jss.2024.07.005>.

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